SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning ***(BL 5)***

Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:100**

Annexure A

Industry Problem Solving

Code of Conduct:

*I **Gayathri N P(192421362)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1, A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

Answer :

Industry Problem

A retail company wants to recommend products to customers based on their past purchasing behaviour. The system should identify customers with similar purchasing patterns and recommend products that those similar customers have purchased.

Proposed Machine Learning Solution

The company can use the K-Nearest Neighbour (KNN) algorithm for product recommendation. KNN is a similarity-based machine learning technique that identifies the most similar customers or products from historical data.

Data Used

The system can consider features such as:

- Products purchased
- Purchase frequency
- Total spending
- Product categories
- Average purchase value
- Customer ratings
- Purchase recency

Similarity Measurement

Similarity between customers can be measured using Euclidean distance or Cosine similarity.

For numerical customer features, Euclidean distance can be calculated as:

$$d(x,y)=\sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$$

A smaller distance means that two customers have more similar purchasing behaviour.

For example, if Customer A and Customer B purchase similar products with similar frequency and spending, their distance will be small.

Recommendation Generation

The recommendation process works as follows:

1. Collect historical customer purchase data.
2. Convert each customer's purchasing behaviour into numerical features.
3. Select a value for K.
4. Calculate the distance between the target customer and other customers.
5. Select the K most similar customers.
6. Identify products purchased by these neighbours.
7. Remove products that the target customer has already purchased.
8. Recommend the most frequently purchased or highly rated remaining products.

For example, if a customer frequently purchases smartphones and similar customers also purchase wireless earphones, phone cases, and chargers, these products can be recommended.

Adaptation to New Customer Data

KNN is an instance-based learning algorithm, so new customer information can be added directly to the existing dataset. When a customer makes a new purchase, their profile is updated. Future recommendations can then be generated using the updated purchasing history.

The system does not require complete retraining whenever a new customer or purchase is added.

Advantages

- Provides personalized recommendations.
- Simple to implement.
- Uses actual customer behaviour.
- Easily incorporates new customer data.
- Can improve customer engagement and sales.

Conclusion

KNN can provide an effective retail recommendation system by identifying customers with similar purchasing behaviour. By continuously adding new purchase information, the system can adapt to changing customer preferences and generate more relevant product recommendations.

2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

Answer:

Industry Problem

A manufacturing company needs to predict machine output using sensor readings such as temperature, pressure, vibration, and speed.

Proposed Solution

Locally Weighted Regression (LWR) can be used because it gives more importance to data points that are closer to the current sensor readings.

Working

1. Collect historical sensor data and machine output.
2. Calculate the distance between the current reading and past readings.
3. Assign higher weights to nearby data points.
4. Build a local regression model.
5. Predict the machine output.

A Gaussian kernel can be used:

$$
w_i=e^{-\frac{(x-x_i)^2}{2\tau^2}}
$$

Improving Accuracy

LWR focuses on similar operating conditions instead of using one global model. Therefore, it can capture local changes in machine behaviour and improve prediction accuracy.

Handling Noisy and Incomplete Data

- **Noise:** Use smoothing, normalization, and outlier removal.
- **Missing data:** Use interpolation, mean/median values, or previous valid readings.

Benefits

- Adapts to changing machine conditions.
- Improves prediction accuracy.
- Useful for real-time monitoring.
- Supports predictive maintenance.

Conclusion

LWR is suitable for manufacturing because it uses nearby historical sensor readings to make accurate machine-output predictions while adapting to changing operating conditions.

3. A healthcare company aims to classify patients into risk categories using historical medical data. Suggest a model using KNN or RBF networks. Explain how the algorithm handles large datasets, feature selection, and accuracy improvement in real-world scenarios.

Answer

Industry Problem

A healthcare company wants to classify patients into **Low, Medium, and High-risk categories** using historical medical data.

Proposed Solution

K-Nearest Neighbour (KNN) can classify a new patient by comparing their medical features with similar patients in the historical dataset.

Working

1. Collect patient data such as age, blood pressure, glucose, cholesterol, and BMI.
2. Clean and normalize the data.
3. Calculate the distance between the new patient and existing patients.
4. Select the **K nearest patients**.
5. Assign the risk category based on majority voting.

Handling Large Datasets

For large datasets, efficient search methods such as **KD-Trees, Ball Trees, or approximate nearest-neighbour techniques** can reduce computation time.

Feature Selection

Irrelevant features can be removed, while important features such as blood pressure, glucose, and cholesterol can be selected. This reduces noise and improves performance.

Accuracy Improvement

Accuracy can be improved using:

- Proper selection of K
- Feature normalization
- Removing outliers
- Cross-validation
- Handling class imbalance

Benefits

- Simple and easy to understand.

- Uses historical patient cases.
- Can be updated with new patient data.
- Supports healthcare risk assessment.

Conclusion

KNN can effectively classify patients into risk categories by comparing them with similar historical cases. Proper feature selection and preprocessing can improve its performance in real-world healthcare applications.

4. An e-commerce platform wants to detect similar products for dynamic pricing strategies. Design a solution using Case-Based Learning. Discuss how similarity metrics are defined, how past cases are reused, and how the system scales with increasing product data.

Answer :

Industry Problem

An e-commerce platform wants to identify similar products to support **dynamic pricing** based on product features, demand, and previous sales.

Proposed Solution

Case-Based Learning (CBL) can be used to compare a new product with previously stored product cases and reuse the information from similar products.

Similarity Measurement

Products can be compared using features such as:

- Product category
- Brand
- Price
- Specifications
- Customer rating
- Sales and demand

Similarity can be calculated using **Euclidean distance** for numerical features and matching methods for categorical features.

Reusing Past Cases

1. Store previous product cases with their pricing and sales information.
2. Extract features of the new product.
3. Find similar products from the case database.
4. Study their previous prices and sales.

5. Use this information to support the new product's pricing decision.
6. Store the new product and its results as a future case.

Scalability

When product data increases, the system can use:

- Product categories
- Feature indexing
- Clustering
- Approximate nearest-neighbour search

These methods reduce the time required to search through all products.

Benefits

- Reuses previous business knowledge.
- Supports dynamic pricing.
- Adapts to new products.
- Reduces decision-making time.
- Improves pricing decisions.

Conclusion

Case-Based Learning is suitable for e-commerce because it identifies similar products and reuses previous pricing experiences to make better decisions for new products.

5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

Answer :

Industry Problem

A logistics company needs to predict delivery time using **location, distance, traffic, weather, and time of day**.

Proposed Solution

A **Radial Basis Function (RBF) Network** can be used because it can learn complex and non-linear relationships between these factors and delivery time.

Working

1. Collect historical delivery data.
2. Use features such as distance, location, traffic, and weather.

3. Normalize the input data.
4. Train the RBF network using previous delivery records.
5. Predict delivery time for new delivery conditions.

The RBF function can be represented as:

[

\phi(x)=e^{-\frac{||x-c||^2}{2\sigma^2}}

]

where (c) is the centre and (\sigma) controls the spread.

Handling Non-Linear Relationships

Delivery time does not always increase directly with distance. For example, a short-distance delivery during heavy traffic may take longer than a long-distance delivery with clear traffic. RBF networks can learn these complex patterns.

Improving Reliability

Prediction reliability can be improved by:

- Using real-time traffic data.
- Removing incorrect or extreme data.
- Normalizing features.
- Regularly updating the model with new delivery data.
- Using cross-validation.

Benefits

- Captures non-linear relationships.
- Provides accurate delivery-time estimates.
- Adapts to changing traffic conditions.
- Helps improve route and delivery planning.
- Increases customer satisfaction.

Conclusion

An RBF network is suitable for logistics because it can learn complex relationships between traffic, location, weather, and delivery time, resulting in more reliable delivery predictions under changing conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand